Monthly Intelligence Report

Edition 001 — Alpine Hydropower & Grid Transition

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

947

147 HV (≥220 kV) · 776 MV · 24 junction

POWER LINES MAPPED

625

~8,000 km coverage · real OSM data

MC SIMULATIONS

9.47 M

10,000 per substation

PUBLIC DATA SOURCES

25

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score. Its engine processes 346,000+ source data points per run, executes 9.47 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and produces full uncertainty quantification across 947 real OSM substations spanning 160 Bezirke, 26 Cantons, and 625 power lines covering approximately 8,000 km.

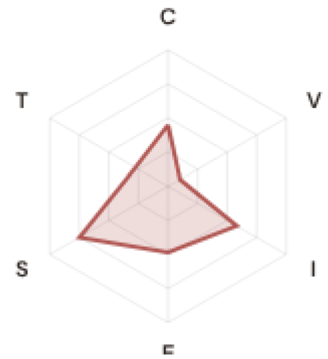
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 25 verified public sources — including BFE, ElCom, Swissgrid, Pronovo, BFS, SED, BAFU, Swisstopo, MeteoSwiss, and OpenStreetMap — making the methodology fully auditable and reproducible. Switzerland presents unique grid challenges: approximately 60% of electricity generated from Alpine hydropower, a multi-decade nuclear phase-out under the Energy Strategy 2050, distributed solar integration across diverse topography, and cross-border energy trading at the heart of the European interconnected grid.

Substation in Focus

Eichli

Lenzburg, Aargau · 16 kV

0.484	Medium	0.152	66th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.448 / 0.30 (149%)

I Infrastructure: 0.582 / 0.25 (233%)

S Saturation: 0.750 / 0.20 (375%)

V Voltage: 0.101 / 0.10 (101%)

E Economic: 0.485 / 0.10 (485%)

T Transition: 0.273 / 0.05 (546%)

MODIFIERS

R3 Consequence: 0.912 ↓ dampens

R4 Graph Criticality: 1.045 ↑ amplifies

R6a Restoration: 1.009 → neutral

R6b Seismic/Network: 1.009 → neutral

R7 Digital Readiness: 1.000 → neutral

This month's spotlight — automatically selected for April 2026 — is **Eichli** in Lenzburg, Aargau. With an R_final of 0.484 (Medium band, 66th fleet percentile), its risk profile is dominated by the T (Transition) component at 546% of its weight ceiling, followed by E (Economic) at 485%. This substation reflects critical transition challenges as Switzerland phases out nuclear capacity while balancing hydropower constraints and renewable integration.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Switzerland's grid faces unique vulnerabilities: Alpine terrain isolation limits redundancy, cross-border energy flows create dependency on neighbours, the nuclear phase-out reduces baseload capacity, and a small population means high per-capita economic consequence per substation failure. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross silos — identifying substations where low digital resilience intersects with high economic consequence.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

0

High/Critical + R7 < median

AVG R7 MODIFIER

1.0000

Range [0.99, 1.05]

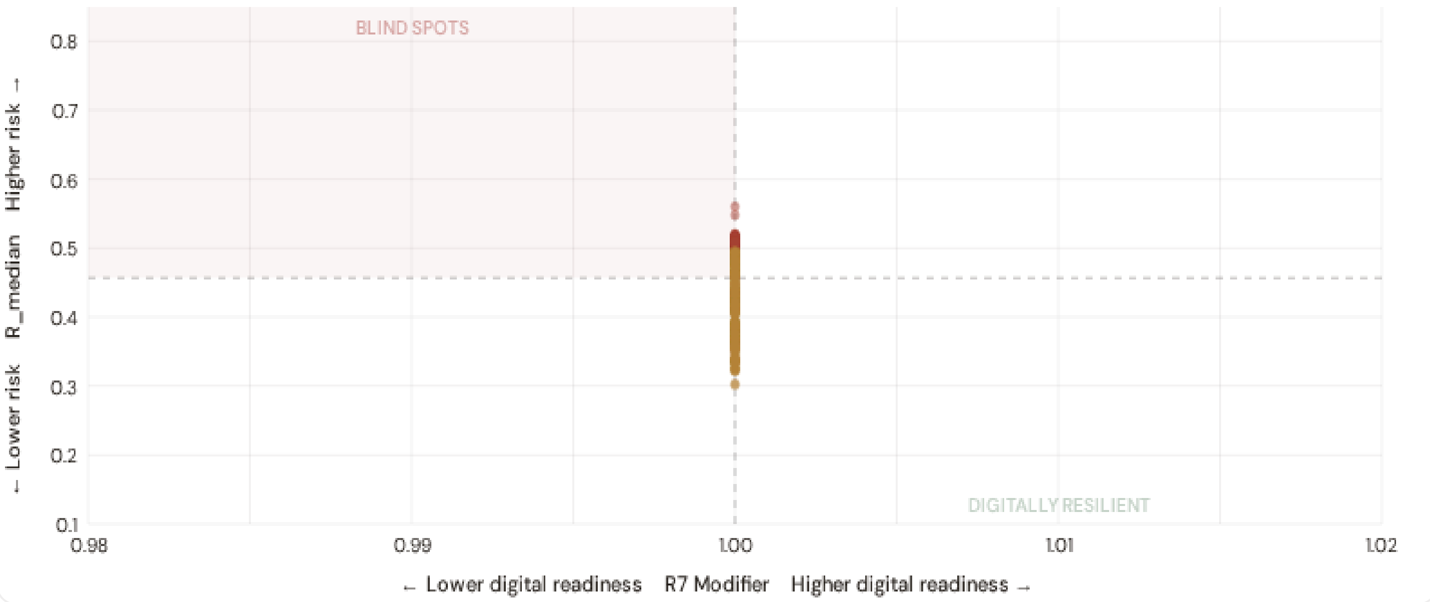
HIGH-CONSEQUENCE SUBS

160

16.9% · R3 ≥ 0.94

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0000). These substations are the most vulnerable to compound events: grid disturbances at locations where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in . Across the full fleet, 0 substations (0.0%) carry HIGH cyber-exposure, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure, concentrated in the Swiss Mittelland urban corridors.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 **Capital-Intensive / High-Consequence**

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

242 substations (25.6%) High/Critical: **146** (60.3%) Avg R: **0.499** Avg E: **0.543** / 0.10

Low 0 Med 96 High 146 Crit 0

🔧 **Industrial / Medium-Large Enterprise**

Est. VoLL: Est. medium-high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

234 substations (24.7%) High/Critical: **25** (10.7%) Avg R: **0.472** Avg E: **0.500** / 0.10

Low 0 Med 209 High 25 Crit 0

🏢 **Commercial / SME-Dense**

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

235 substations (24.8%) High/Critical: **0** (0.0%) Avg R: **0.422** Avg E: **0.500** / 0.10

Low 0 Med 235 High 0 Crit 0

🌾 **Light / Agricultural / Remote**

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

236 substations (24.9%) High/Critical: **0** (0.0%) Avg R: **0.381** Avg E: **0.488** / 0.10

Low 0 Med 236 High 0 Crit 0

Value of Lost Load (VoLL) context: ElCom estimates Swiss average VoLL at CHF 16–25/kWh for commercial customers and CHF 4–8/kWh for residential. Switzerland's high GDP per capita (CHF 87,000+) means per-substation economic exposure is among the highest in Europe. The SSI's R3 consequence multiplier identifies which substations serve the most economically exposed territories. **111 substations** combine high-consequence economic fabric (R3 ≥ 0.94) with High or Critical risk classification — the highest VoLL-weighted exposure in the fleet.

The financial and pharmaceutical tier concentrates in **Zürich (193), Bern (41), Luzern (4)** — Cantons where banking, precision manufacturing, and pharmaceutical industries depend on uninterrupted power for trading systems, clean rooms, and continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of CHF 20–40/kWh, compared to CHF 1–4/kWh in Alpine rural areas. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores at this granularity.

B.3 Bezirk Digital Readiness Ranking

Bezirke ranked by average R7 modifier — lower values indicate weaker digital infrastructure. Cross-referenced with average R_median to identify Bezirke combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 BEZIRKE

Longer bar = higher R7 = better digital infrastructure.

1	Leventina	1.0000
2	Hochdorf	1.0000
3	Fricktal ⚠ high R	1.0000
4	Ilanz	1.0000
5	Brugg ⚠ high R	1.0000
6	Renens ⚠ high R	1.0000
7	Delemont	1.0000
8	Winterthur ⚠ high R	1.0000
9	Pinot ⚠ high R	1.0000
10	Bernina	1.0000
11	Albula	1.0000
12	Surselva	1.0000
13	Sierre	1.0000
14	Conthey	1.0000
15	Martigny	1.0000

Bezirk-level analysis reveals geographic variation in digital infrastructure that mirrors Alpine topography constraints. **Leventina** has the lowest average R7 (1.0000), while **Franches-Montagnes** leads at 1.0000 — a spread of 0.0000 across the R7 range. **0 Bezirke** combine below-median digital readiness with above-median grid risk, making them priority targets for Swisscom fibre and smart-grid investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open data — as NIS2 transposition proceeds and ElCom enhances its digital monitoring requirements, future editions will expand R7's dynamic range.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0000, median = 1.0000; 0 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging Bezirke where smart-grid deployments and broadband expansion translate into measurable R7 shifts.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **25 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. All sources are publicly accessible — from BFE and ElCom reporting through to OSM infrastructure topology and SED seismic hazard data.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL
25	5	14

95 variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

6

Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
SWGRID	Swissgrid	MONTHLY	TSO zone	3 vars
PRON	Pronovo	MONTHLY	Canton	4 vars
NABEL	NABEL Air Monitoring Network	CONTINUOUS	Station	2 vars
SNB	SNB (Swiss National Bank)	QUARTERLY	National	2 vars
BFE	BFE (Swiss Federal Office of Energy)	ANNUAL	Canton	6 vars
ELCOM	ElCom (Electricity Commission)	ANNUAL	Canton	5 vars
BFS	BFS (Federal Statistical Office)	ANNUAL	Bezirk	8 vars
BAFU	BAFU (Federal Office for Environment)	ANNUAL	Canton / station	4 vars
SECO	SECO (State Secretariat for Economy)	ANNUAL	Canton	3 vars
ASTRA	ASTRA (Federal Roads Office)	ANNUAL	Canton	1 var
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
WEKO	WEKO (Competition Commission)	ANNUAL	Canton	1 var
ARE	ARE (Spatial Development Office)	ANNUAL	Bezirk	1 var
SED	SED (Swiss Seismological Service)	STATIC	~5 km	2 vars
STOPO	Swisstopo	STATIC	Bezirk	2 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
ISO-9223	ISO 9223 Corrosion	DERIVED	Bezirk	1 var
METEO	MeteoSwiss	DAILY	~1 km	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Fleet Score Snapshot

FLEET SIZE

947

947 real OSM substations

MEDIAN R

0.457

Mean: 0.443

HIGH + CRITICAL

171

18.1% of fleet

LOW BAND

0

0.0% of fleet

No primary data sources have been updated since the v4.0.2 launch baseline. All **947 substation scores remain unchanged** this edition. The fleet median R of 0.457 (mean 0.443) reflects a distribution with 0.0% in Low and 81.9% in Medium. The first material score changes are expected when ElCom publishes updated SAIDI/SAIFI statistics (affecting C1–C4 and R6a) and when Pronovo refreshes the renewable generation registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

SECTION D — MONTHLY DEEP-DIVE

Alpine Hydropower & Grid Transition: Where Geography Meets the Energy Transition

> **Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Alpine hydropower & grid transition · **002** Nuclear phase-out and baseload replacement · **003** DER saturation in the Swiss Mittelland · **004** Cross-border energy trading and interconnector risk · **005** Voltage quality in Alpine valleys · **006** Infrastructure age vs cantonal investment patterns · **007** Seismic hazard along the Alpine front · **008** T-component forward projections (Energy Strategy 2050+) · **009** Cantonal Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication progress · **012** Year in review.

D.1 Why Alpine Hydropower, Why Now

HYDROPOWER SHARE

~60%

Of Swiss electricity generation

ALPINE CANTONS

9

272 substations in Alpine corridor

ALPINE CORRIDOR MEDIAN R

0.420

Fleet median: 0.457

HIGHEST-RISK CANTON

Bern

Med R: 0.501 · 77 subs

Switzerland generates approximately 60% of its electricity from hydropower, with the vast majority concentrated in Alpine cantons where steep gradients and reservoir capacity provide both storage and generation. The SSI captures **272 substations** across 9 Alpine cantons. Their corridor median R of 0.420 sits at the fleet median of 0.457, with 28% of Alpine substations scoring above fleet median. **42 are classified High**. The combination of remote Alpine locations, long transmission distances, seismic exposure along the Alpine front, and seasonal demand fluctuation from tourism creates a unique stress profile that single-metric indices cannot capture.

D.2 Alpine Corridor Scoring

CANTON	SUBS	MED R	HIGH	C	I	S	T
Bern	77	0.501	42	0.448	0.577	0.781	0.259
Ticino	32	0.434	0	0.447	0.660	0.354	0.104
Valais	64	0.420	0	0.448	0.586	0.333	0.089
Graubünden	50	0.380	0	0.448	0.580	0.302	0.053
Glarus	1	0.363	0	0.448	0.637	0.177	0.070
Schwyz	17	0.357	0	0.447	0.594	0.229	0.066
Uri	24	0.337	0	0.447	0.618	0.156	0.029
Obwalden	4	0.326	0	0.448	0.624	0.117	0.035
Nidwalden	3	0.326	0	0.448	0.626	0.112	0.033

D.3 Component Analysis — Alpine vs Mittelland

ALPINE CANTONS (AVG)

C Continuity	0.448
I Infrastructure	0.596
S Saturation	0.428
V Voltage	0.114
E Economic	0.487
T Transition	0.124

MITTELLAND & URBAN CANTONS (AVG)

C Continuity	0.447
I Infrastructure	0.578
S Saturation	0.578
V Voltage	0.097
E Economic	0.516
T Transition	0.228

The component-level comparison reveals where Alpine grid stress diverges from the Swiss Mittelland. The largest gap is in **S Saturation** (-0.149 for Alpine), followed by **T Transition** (-0.104). Alpine substations face higher infrastructure stress (longer lines, older equipment, weather exposure) and higher saturation pressure from seasonal tourism peaks, while Mittelland substations show higher economic consequence density due to their proximity to financial and industrial centres.

D.4 Implications for the Energy Strategy 2050+

The Energy Strategy 2050+ sets ambitious targets: full nuclear phase-out by 2044, net-zero emissions by 2050, and a doubling of renewable capacity. For the Alpine corridor, this creates three compounding pressures. First, the loss of baseload nuclear generation (currently ~35% of supply) increases reliance on Alpine hydropower for load-balancing, placing additional stress on mountain transmission infrastructure. Second, new solar and wind installations in Alpine valleys require grid reinforcement in precisely the locations where terrain makes construction most difficult and expensive. Third, climate change is altering precipitation patterns and glacier melt timing, introducing new uncertainty into reservoir management. The SSI captures these overlapping stresses through its T-component (transition pressure), I-component (infrastructure condition), and R6b seismic modifier — providing a composite view that single-domain analyses cannot match. For Swissgrid's Strategic Grid 2040 planning, the SSI identifies priority reinforcement corridors where multiple risk dimensions converge.

SECTION E

European Context Card

Switzerland in the European grid: Despite not being an EU member, Switzerland is electrically embedded at the heart of the European interconnected system. Its grid connects to France, Germany, Austria, and Italy through 41 cross-border transmission lines, making it a critical transit corridor for north-south power flows. This integration creates both resilience (import capacity during domestic shortfalls) and vulnerability (exposure to continental frequency events and cascading failures).

CROSS-BORDER LINES

41

To FR, DE, AT, IT

NET IMPORT (TYPICAL)

~5 TWh

Winter dependence on neighbours

SSI COUNTRIES LIVE

4

IT · DE · AT · CH

ENTSO-E ZONE

CH

Synchronous Continental Europe

Switzerland's position as a European transit hub means that domestic grid stress cannot be assessed in isolation. The SSI v4.0.2 captures cross-border dependency through several channels: the R3 consequence multiplier reflects the economic importance of interconnection points, the R4 graph criticality modifier identifies substations whose failure would disrupt transit flows, and the E component incorporates import/export dependency into regional economic exposure. With four countries now live in the SSI — Italy, Germany, Austria, and Switzerland — cross-border analysis will become increasingly possible in future editions.

Key European developments relevant to Swiss grid resilience this quarter include: the EU's revised Network Code on Cybersecurity (which Switzerland is expected to adopt bilaterally), ongoing negotiations on a new electricity agreement with the EU, and Swissgrid's participation in the ENTSO-E Winter Outlook process. The absence of a formal EU electricity agreement creates regulatory uncertainty that the SSI's confidence intervals partially capture through wider CI bands in border regions.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DNO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a CHF 10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against ElCom's published reliability data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Swiss fleet comprises 947 substations with complete 95-feature vectors from 25 public data sources. Average CI_width of 0.154 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 776 Medium, 171 High, 0 Critical — provides training signal across risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

Valais Hydro-Solar Nexus

Deep-dive into Valais's grid risk profile — substation map, component analysis, province breakdown. European Context Card: undefined.

NEXT DATA REFRESH

Q3 2026 — 1 Quarterly Sources

SNB are due for refresh in Q3. Impact: DER registry updates (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **BFS (Federal Statistical Office)** (8 variables → R3 population, elderly, GDP, fiscal capacity, demographics).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Valais**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Swiss utility, cantonal energy office, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

Published 9 April 2026 · SSI Index v4.0.2 · 947 substations · 625 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5